# Conformal Invariance and Sharp Wilson Majorant Boundary Bounds

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## Abstract

For the scalarized positive Wilson majorant, we establish an exact mass-independent weighted Fourier identity and a uniform bound on finite-interval Hilbert–Schmidt boundary leakage. The proof identifies the majorant symbol at each positive mass parameter with a disk-automorphic composition of one reference symbol and applies conformal invariance of the boundary homogeneous half-Sobolev seminorm. We derive the exact deficit from the saturated boundary energy and its fixed-mass finite-interval asymptotic. For the fixed-scaled-mass continuum leakage function, we determine its exact global supremum and its two leading inverse-power corrections at large scaled mass. The resulting uniform estimate improves the prefactor available in the reference-modular boundary-term bound, but does not settle the longitudinal reference-modular trace or the complete relative-entropy asymptotic.

## 1. Conventions

For a parameter $r > 0$, define the real, even $2\pi$-periodic scalar Wilson-majorant symbol

$$G_r(k) = \frac{1}{2}\left(1 - \frac{|\sin(k/2)|}{\sqrt{\sin^2(k/2) + r^2\cos^2(k/2)}}\right), \qquad -\pi \le k \le \pi,$$

with Fourier coefficients

$$c_d(r) = \frac{1}{2\pi}\int_{-\pi}^{\pi} G_r(k) e^{-idk}\, dk.$$

In the cell-preserving scalarization of the two-orbital lattice, these coefficients occur at scalar displacements $2d$. For a scalar interval of even length $N$ and projection $\chi_N$, the exact full-line Fourier boundary identity is

$$\mathcal{L}_N(r) := \|(I - \chi_N)G_r\chi_N\|_{\mathrm{HS}}^2 = 2\sum_{d=1}^{\infty} \min(N, 2d)|c_d(r)|^2. \tag{1}$$

This identity counts the two directions across the interval boundary and uses the Fourier coefficient normalization above. It is not a statement about an arbitrary two-orbital matrix-valued symbol under a different scalarization.

## 2. Universal weighted Fourier identity

**Theorem 1.** For every $r > 0$,

$$\boxed{4\sum_{d=1}^{\infty} d\,|c_d(r)|^2 = \frac{1}{2\pi^2}.} \tag{2}$$

Consequently, for every admissible interval length $N$,

$$\boxed{0 < \mathcal{L}_N(r) < \frac{1}{2\pi^2},} \tag{3}$$

and its exact positive deficit is

$$\boxed{\frac{1}{2\pi^2} - \mathcal{L}_N(r) = 2 \sum_{d>N/2} (2d - N)|c_d(r)|^2.} \tag{4}$$

*Proof.* Put $a = (1-r)/(1+r) \in (-1,1)$, and let

$$M_a(z) = \frac{z-a}{1-az}, \qquad |z| = 1.$$

Writing $M_a(e^{ik}) = e^{i\psi_r(k)}$, the half-angle identity gives

$$\tan\frac{\psi_r(k)}{2} = \frac{1}{r}\tan\frac{k}{2}.$$

The formula defining $G_r$ therefore gives the pointwise boundary identity

$$G_r(e^{ik}) = G_1(M_a(e^{ik})), \qquad G_1(k) = \tfrac{1}{2}(1 - |\sin(k/2)|). \tag{5}$$

For a real boundary function $u$ with Fourier coefficients $\widehat{u}_n$, define the homogeneous half-Sobolev energy

$$\mathcal{E}(u) = \sum_{n\in\mathbb{Z}} |n|\,|\widehat{u}_n|^2 = \frac{1}{4\pi^2}\int_0^{2\pi}\int_0^{2\pi} \frac{|u(e^{i\theta}) - u(e^{i\varphi})|^2}{|e^{i\theta} - e^{i\varphi}|^2}\, d\theta\, d\varphi. \tag{6}$$

The identity holds first for trigonometric polynomials, by orthogonality and the Dirichlet-kernel identity, and then by approximation in the homogeneous $H^{1/2}$ norm. Its normalization can be checked on $u(e^{ik}) = e^{ik}$. A disk automorphism satisfies

$$|M_a(z) - M_a(w)|^2 = |M_a'(z)|\,|M_a'(w)|\,|z-w|^2, \quad |z| = |w| = 1,$$

and its boundary angular Jacobian equals $|M_a'(z)|$. Changing both variables in (6) proves $\mathcal{E}(u \circ M_a) = \mathcal{E}(u)$. This also follows from conformal invariance of the Dirichlet energy of harmonic extensions. In particular, (5) implies $\mathcal{E}(G_r) = \mathcal{E}(G_1)$.

For $d \geq 1$, elementary integration yields

$$c_d(1) = -\frac{1}{2\pi}\int_0^{\pi} \sin(k/2)\cos(dk)\, dk = \frac{1}{\pi(4d^2-1)}. \tag{7}$$

Since $G_r$ is real and even, $\mathcal{E}(G_r) = 2\sum_{d\geq 1} d|c_d(r)|^2$. The partial sum at $r = 1$ telescopes:

$$4\sum_{d=1}^{D} d|c_d(1)|^2 = \frac{4}{\pi^2}\sum_{d=1}^{D}\frac{d}{(4d^2-1)^2} = \frac{1}{2\pi^2}\left(1 - \frac{1}{(2D+1)^2}\right). \tag{8}$$

Letting $D \to \infty$ and using invariance gives (2). Comparing (1) with the saturated sum (2) proves (3) and (4); strictness follows because the boundary symbol is not a trigonometric polynomial and hence has nonzero Fourier coefficients at arbitrarily large indices. □

## 3. Fixed-parameter finite-interval deficit

**Theorem 2.** For each fixed $r > 0$, as $N \to \infty$ through even integers,

$$\boxed{\mathcal{L}_N(r) = \frac{1}{2\pi^2} - \frac{1}{6\pi^2 r^2 N^2} + o_r(N^{-2}).} \tag{9}$$

*Proof.* Let $h_r(k) = 2G_r(k)$ for $0 \le k \le \pi$. Its one-sided endpoint derivative is $h_r'(0) = -1/(2r)$, while $h_r'(\pi) = 0$, and $h_r'' \in L^1(0,\pi)$ at every fixed $r > 0$. Twice integrating the cosine coefficient by parts, then applying the Riemann–Lebesgue lemma, gives

$$\lim_{d\to\infty} d^2 c_d(r) = \frac{1}{4\pi r}. \tag{10}$$

Using (4) with $A = N/2$, the asymptotic (10) is uniform in the tail $d > A$ as $A \to \infty$, whence

$$\begin{aligned}\frac{1}{2\pi^2} - \mathcal{L}_N(r) &= 4\sum_{d>A}(d-A)|c_d(r)|^2 \\ &= \frac{1+o_r(1)}{4\pi^2 r^2}\sum_{d>A}\frac{d-A}{d^4} \\ &= \frac{1+o_r(1)}{4\pi^2 r^2}\left(\frac{1}{6A^2} + o(A^{-2})\right) = \frac{1}{6\pi^2 r^2 N^2} + o_r(N^{-2}).\end{aligned}$$

The asymptotic is for **fixed** $r$. It must not be used as a uniform approximation when $r$ depends on $N$. □

## 4. Exact continuum saturation and its large-mass expansion

For $\mu > 0$, set $r_N = \mu/(N+\mu)$. Define

$$f(t) = 1 - \frac{t}{\sqrt{1+t^2}}, \qquad F(x) = \int_0^\infty f(t)\cos(2xt)\,dt.$$

The continuum boundary-leakage function is

$$\Phi(\mu) = \frac{2}{\pi^2}\int_0^\infty \min(\mu, 2x)F(x)^2\,dx. \tag{11}$$

This expression is the fixed-scaled-mass limit of $\mathcal{L}_N(r_N)$; its derivation from scaled Fourier coefficients is independent of the conclusions below.

**Theorem 3.** The following weighted transform integral is exact:

$$\boxed{\int_0^\infty xF(x)^2\,dx = \frac{1}{8}.} \tag{12}$$

Consequently,

$$\boxed{\lim_{\mu\to\infty}\Phi(\mu) = \sup_{\mu>0}\Phi(\mu) = \frac{1}{2\pi^2}.} \tag{13}$$

Moreover, as $\mu \to \infty$,

$$\boxed{\Phi(\mu) = \frac{1}{2\pi^2} - \frac{1}{6\pi^2\mu^2} - \frac{3}{10\pi^2\mu^4} + O(\mu^{-6}).} \tag{14}$$

*Proof.* We first give a Fourier-analytic derivation of (12) independent of the Möbius argument. Set $h(t) = (1+t^2)^{-3/2}$ and $H(k) = \int_0^\infty h(t)\sin(kt)\,dt$. Integration by parts yields $F(x) = H(2x)/(2x)$ for $x > 0$. A Laplace regularization of the positive quadratic integral gives

$$\int_0^\infty \frac{H(k)^2}{k}\,dk = \frac{1}{2}\int_0^\infty \int_0^\infty h(t)h(s)\log\frac{t+s}{|t-s|}\,dt\,ds. \tag{15}$$

Indeed, inserting $e^{-\varepsilon k}$, using the elementary sine-product integral, and taking $\varepsilon \downarrow 0$ yields (15); the logarithmic diagonal singularity is locally integrable against the smooth integrable factors. Substitute $t = \tan u$, $s = \tan v$, so that $h(t)dt = \cos u\,du$ and

$$\log\frac{t+s}{|t-s|} = \log\frac{\sin(u+v)}{|\sin(u-v)|}.$$

For $0 < u, v < \pi/2$, the classical logarithmic sine Fourier series, interpreted by Abel summation, gives

$$\log\frac{\sin(u+v)}{|\sin(u-v)|} = 2\sum_{n=1}^{\infty}\frac{\sin(2nu)\sin(2nv)}{n}.$$

The integrated coefficients are

$$a_n = \int_0^{\pi/2}\cos u\sin(2nu)\,du = \frac{2n}{4n^2-1}.$$

Abel regularization and the integrability of the logarithmic kernel justify integrating the Fourier series. Therefore

$$\int_0^\infty \frac{H(k)^2}{k}\,dk = \sum_{n=1}^{\infty}\frac{a_n^2}{n} = \sum_{n=1}^{\infty}\frac{4n}{(4n^2-1)^2} = \frac{1}{2}. \tag{16}$$

As $xF(x)^2\,dx = \frac{1}{4}H(k)^2dk/k$ under $k = 2x$, (12) follows. Monotone convergence in (11) gives (13).

For the refined expansion, integration by parts yields

$$F(x) = \frac{1}{2x}\int_0^\infty h(t)\sin(2xt)\,dt.$$

Since $h(0) = 1$, $h''(0) = -3$, and sufficiently many derivatives of $h$ are integrable, repeated integration by parts establishes

$$F(x) = \frac{1}{4x^2} + \frac{3}{16x^4} + O(x^{-6}). \tag{17}$$

Subtract (11) from (13):

$$\frac{1}{2\pi^2} - \Phi(\mu) = \frac{2}{\pi^2}\int_{\mu/2}^\infty (2x-\mu)F(x)^2\,dx. \tag{18}$$

Substituting $F(x)^2 = 1/(16x^4) + 3/(32x^6) + O(x^{-8})$ into (18), with the remainder uniformly integrable over $x \geq \mu/2$, gives

$$\frac{1}{2\pi^2} - \Phi(\mu) = \frac{1}{6\pi^2\mu^2} + \frac{3}{10\pi^2\mu^4} + O(\mu^{-6}),$$

which proves (14). □

## 5. A uniform modular boundary estimate

Let $P_0$ be the full-line half-filled reference projector and $B_N = \chi_N P_0 \chi_N$, $W_N = B_N(I - B_N)$. For any self-adjoint finite reference-modular discrepancy $E_N$, put

$$C_N = \chi_N G_r (I - \chi_N) P_0 \chi_N.$$

The Hilbert–Schmidt Cauchy–Schwarz inequality and the full-line projector identity give

$$2|\operatorname{Tr}(C_N E_N)| \le 2\sqrt{\mathcal{L}_N(r)}\,\sqrt{\operatorname{Tr}(W_N E_N^2)}.$$

Theorem 1 yields the **uniform** inequality

$$\boxed{2|\operatorname{Tr}(C_N E_N)| \le \frac{\sqrt{2}}{\pi}\,\sqrt{\operatorname{Tr}(W_N E_N^2)}.} \tag{19}$$

This coefficient does not depend on $r$ or $N$. Under the separately established estimate $\operatorname{Tr}(W_N E_N^2) = O(\log N)$, the right-hand side remains $O(\sqrt{\log N})$. It neither establishes $\operatorname{Tr}(S_N E_N) = o(\log N)$ nor proves the leading logarithmic coefficient or finite part of the complete Wilson relative entropy.

## 6. Mathematical scope

The exact identity (2) uses the stated scalar symbol and even-displacement scalarization. The fixed-$r$ asymptotic (9) and fixed-$\mu$ continuum limit (11) are different limiting operations; (9) is not uniform in a mass parameter shrinking with $N$. The large-$\mu$ expansion (14) concerns the continuum function after the finite-interval limit. None of these results addresses the joint sine-kernel/Jacobi moving-index spectrum required for the longitudinal modular estimate.

The separate weighted argument in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law* establishes the complete leading law with bounded remainder. The local non-implication statements above describe the scope of the conformal method; they do not leave that leading coefficient unresolved.

## References


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